Training runs: ANP with BGE-M3 item embeddings (cross-attention, 1 self-at

Same architecture throughout; the split is the variable. Replay: random acquisition, macro-task Brier ALC.

run	split	width / latent	context sizes, method	steps	best step	bank Brier	replay val	replay test
ANP, random-subject split	random subject	64 / 32	uniform, 0-31 labels	100,000	79,700	0.1248	0.1519	0.2465
ANP, random-subject split (width 128)	random subject	128 / 128	uniform, 0-31 labels	100,000	97,800	0.1193	0.1485	0.2515
ANP	benchmark holdout	64 / 32	uniform, 0-31 labels	100,000	1,200	0.1675	0.1735	0.2725
ANP, budget-ladder training	benchmark holdout	64 / 32	budget ladder, ALC weights	15,000	50	0.2105	not run	not run
ANP + SAM	benchmark holdout	64 / 32	budget ladder, equal weights, SAM 0.05	30,000	5,450	0.2058	0.1667	0.2663
ANP + prior anchor	benchmark holdout	64 / 32	budget ladder, equal weights, anchor	10,000	100	0.2066	not run	not run
ANP + SAM, budget-ladder training	benchmark holdout	64 / 32	budget ladder, ALC weights, SAM 0.05	30,000	4,350	0.2063	not run	not run
ANP, dense-prefix training	benchmark holdout	64 / 32	every prefix, 0-31	10,000	7,800	0.1694	0.1735	0.2762

bank Brier: best value on the run's own fixed validation bank, the metric that picked its checkpoint. That bank's context sizes follow the run's sampling scheme, and in-distribution banks hold benchmarks the run trained on, so bank Brier compares only runs with the same bank.

replay val / test: random acquisition on the 82-task validation and 45-task test panels, the same protocol for every row. The random-split runs trained on ai2d_test and bird_sql (validation) and editbench (test).

The dense-prefix run kept training after its replay and CodaBench checkpoint was frozen at step 4,550 (bank Brier 0.1700); its final best is the step shown. SAM ran to 30,000 steps but its best checkpoint, step 5,450, did not change.

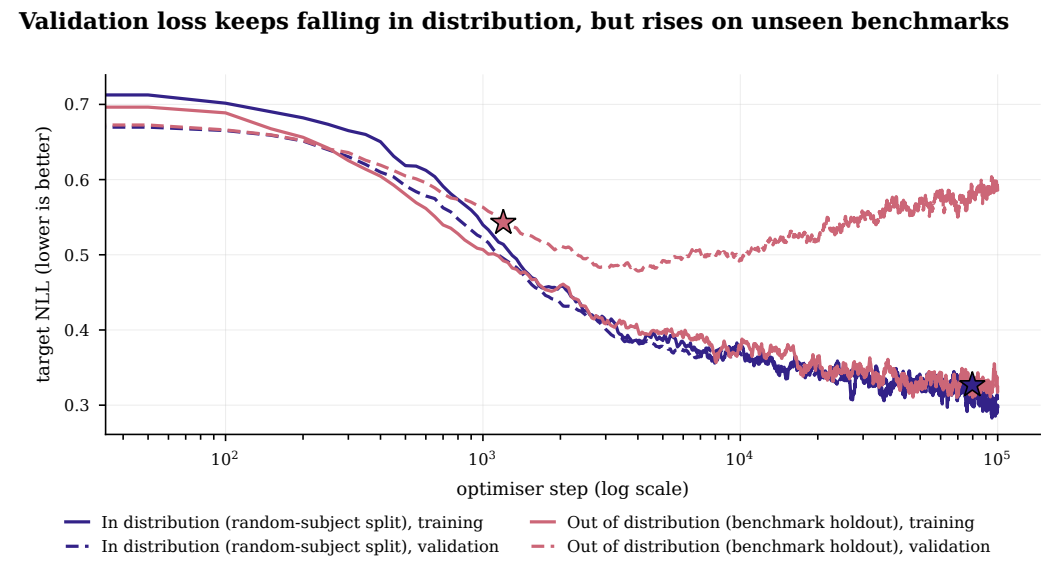
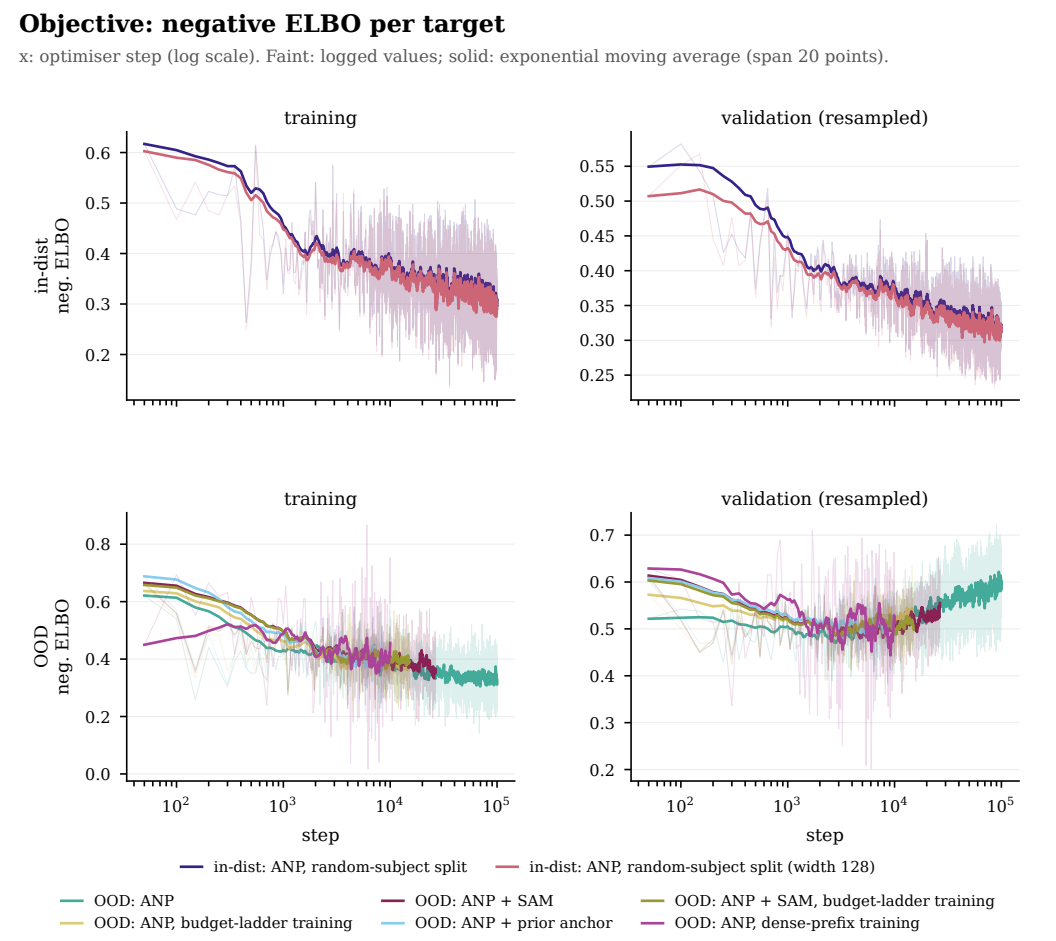
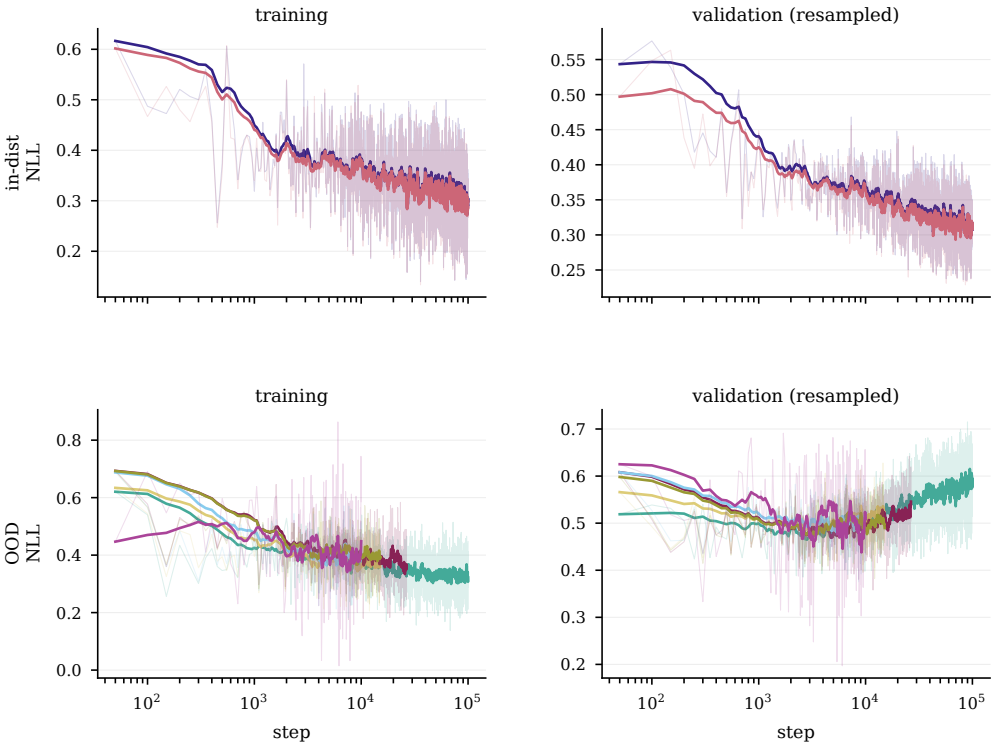


Figure. Training (solid) and validation (dashed) negative log-likelihood of the same ANP (width 64, uniform 0-31 context sizes, 100,000 steps) trained on two splits; curves are exponentially smoothed and stars mark each run's selected checkpoint. In distribution, validation holds out subjects but keeps every benchmark in training, and its loss falls with the training loss throughout. Under the benchmark holdout, validation uses benchmarks never trained on: its loss bottoms out near 0.48 around 3,000 steps and climbs to about 0.59 while the training loss keeps falling, the signature of fitting the training benchmarks rather than learning what transfers. The two validation sets differ by design, so compare each curve's shape, not their levels.


Objective: negative ELBO per target
x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).
training
validation (resampled)
in-dist
neg. ELBO
OOD
neg. ELBO
step
in-dist: ANP, random-subject split
in-dist: ANP, random-subject split (width 128)
OOD: ANP
OOD: ANP + SAM
OOD: ANP + SAM, budget-ladder training
OOD: ANP, budget-ladder training
OOD: ANP + prior anchor
OOD: ANP, dense-prefix training

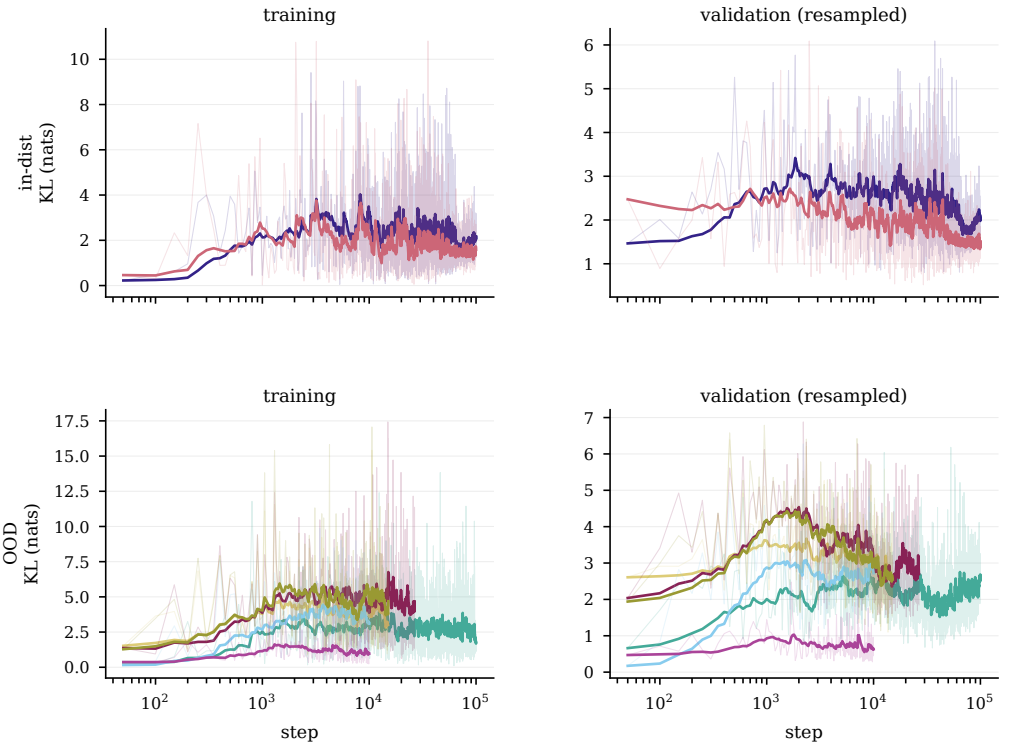
Posterior target NLL

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



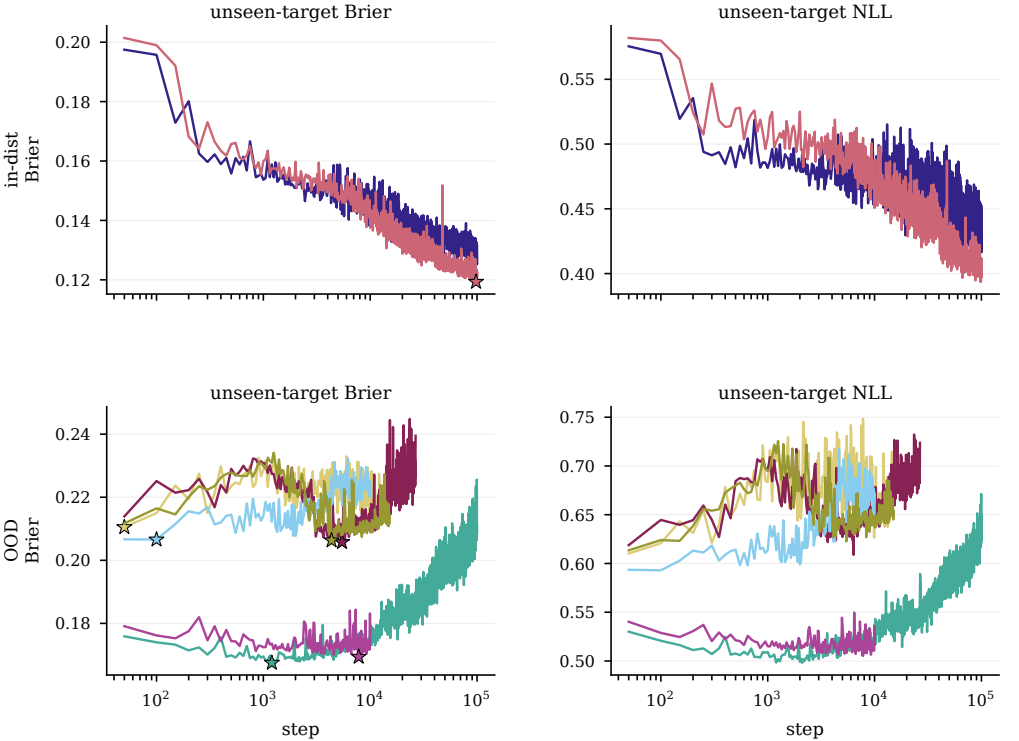
Latent KL (total)

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



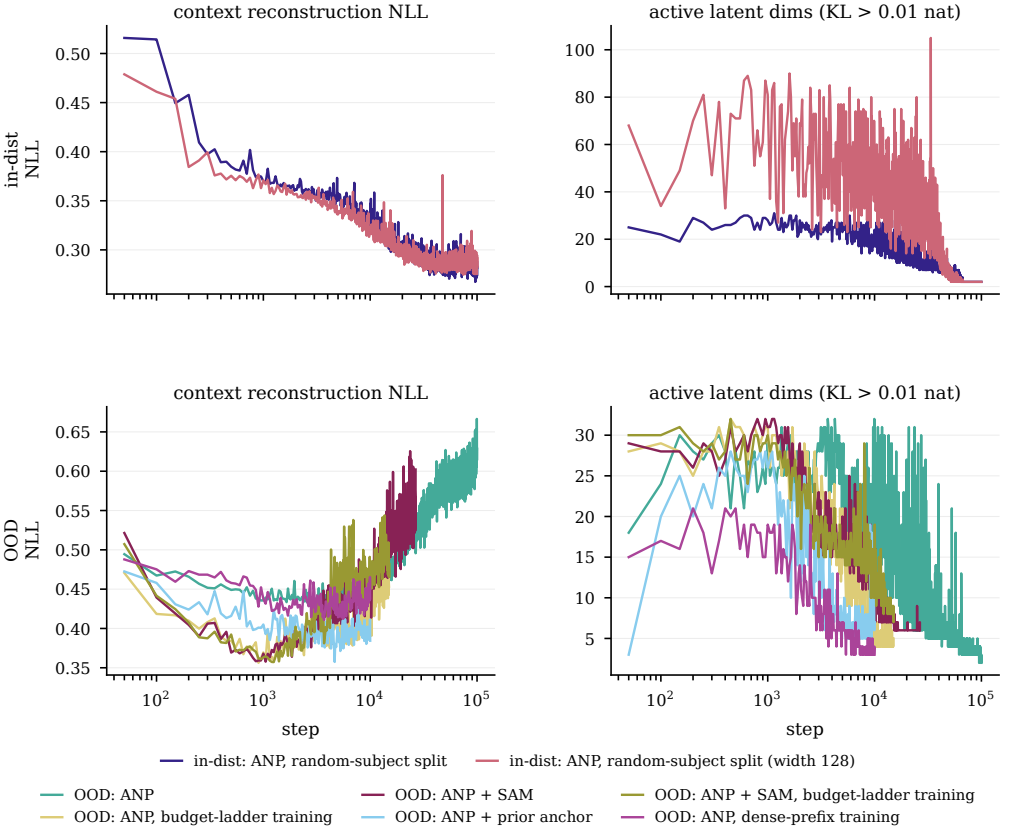
Fixed validation bank: checkpoint-selection metric

x: optimiser step (log scale). Star: the step each run's checkpoint was selected at.



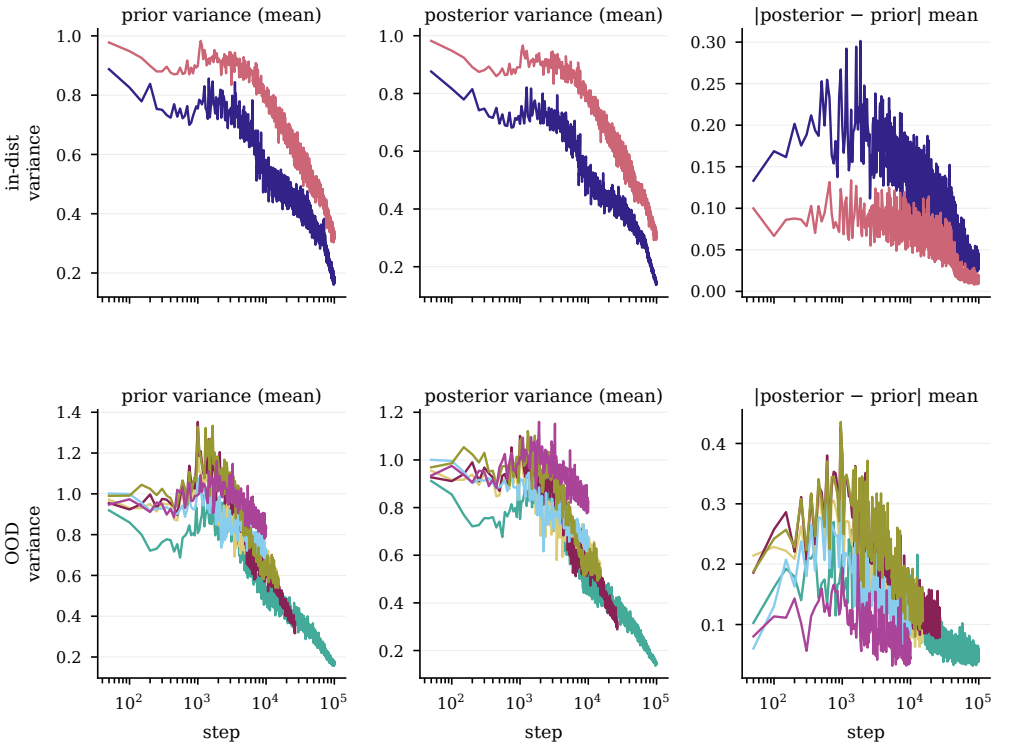
Fixed validation bank: context fit and latent use

x: optimiser step (log scale). Values as logged.



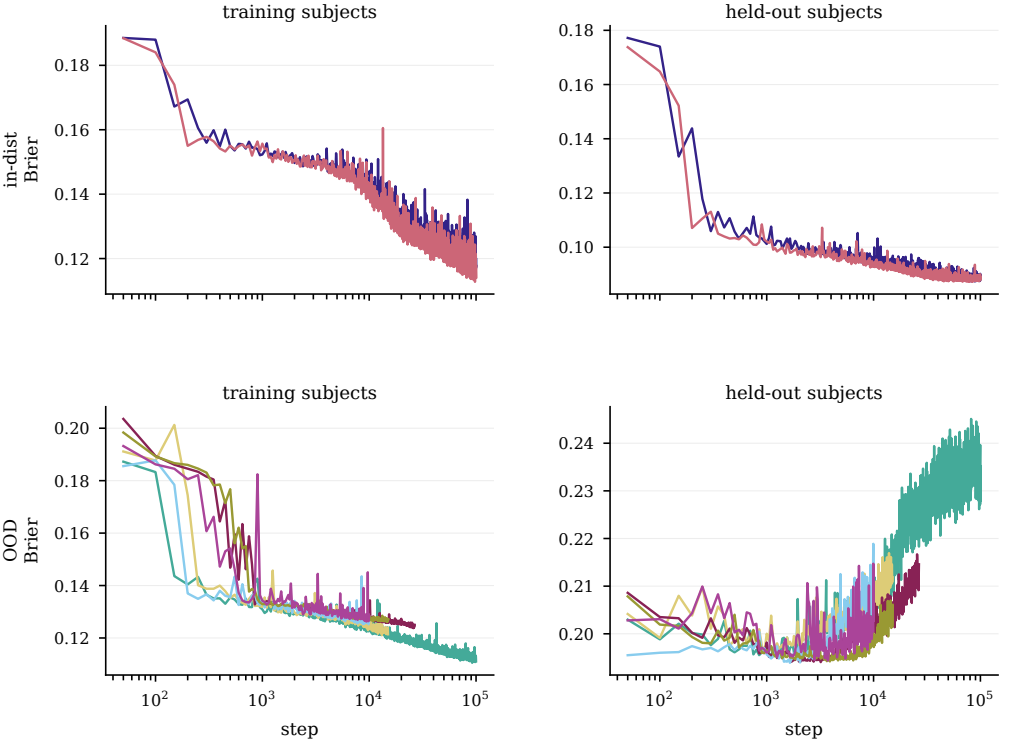
Latent distribution

x: optimiser step (log scale). Values as logged.



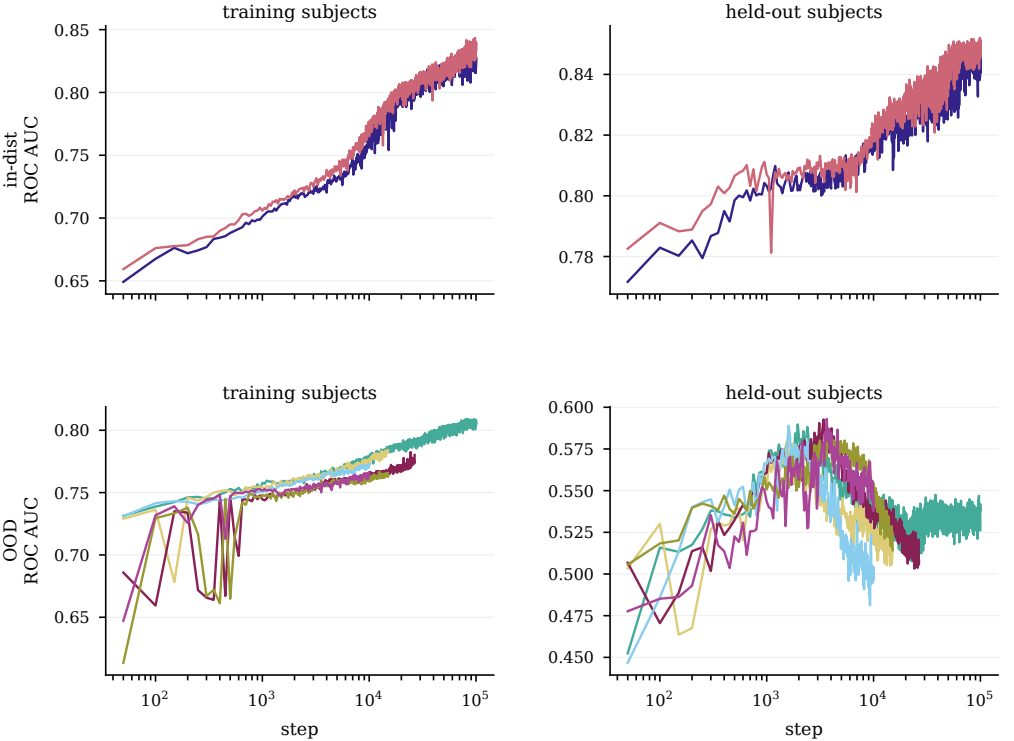
Generalisation: same-subject split-half Brier

x: optimiser step (log scale). Values as logged.



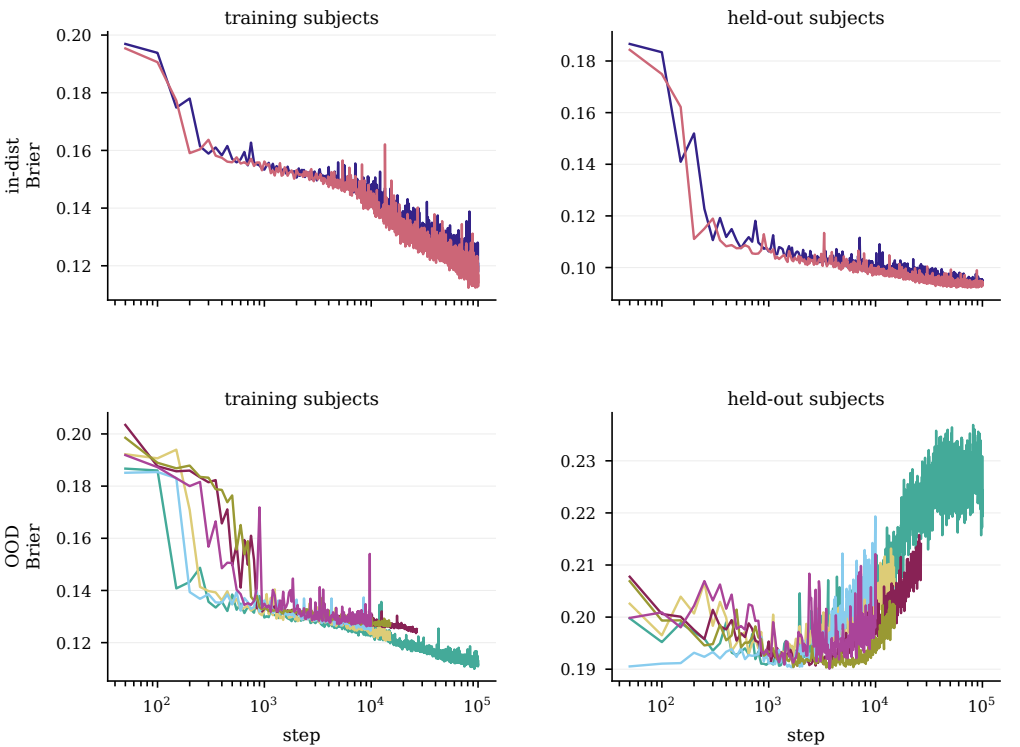
Generalisation: same-subject split-half AUC

x: optimiser step (log scale). Values as logged.



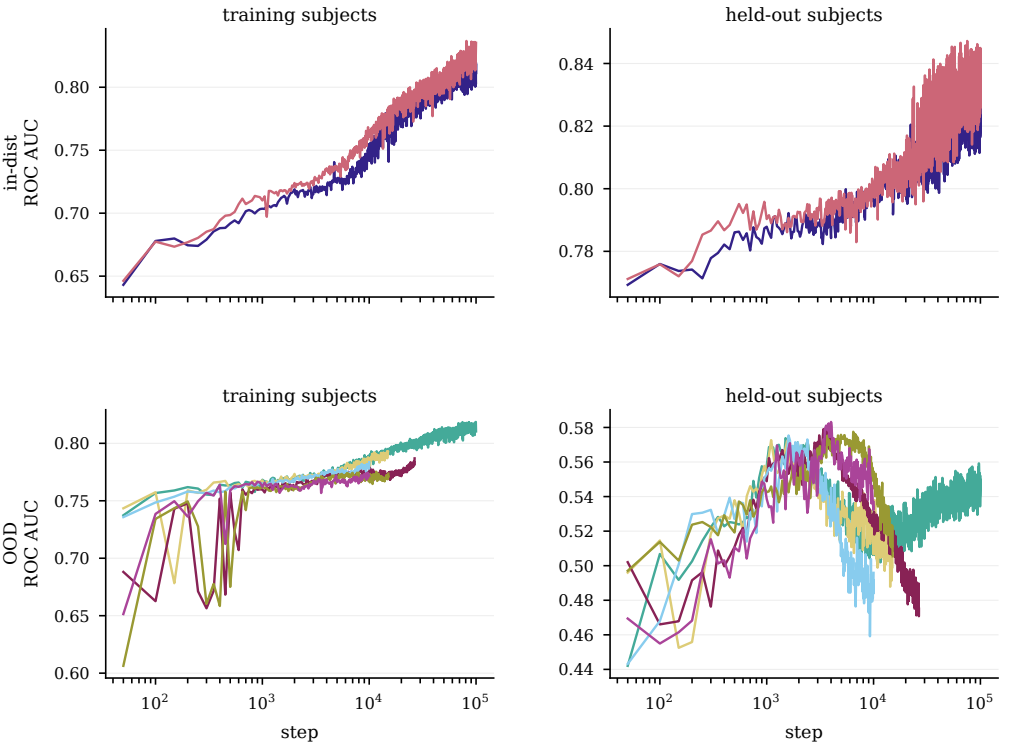
Generalisation: same-subject response-holdout Brier

x: optimiser step (log scale). Values as logged.



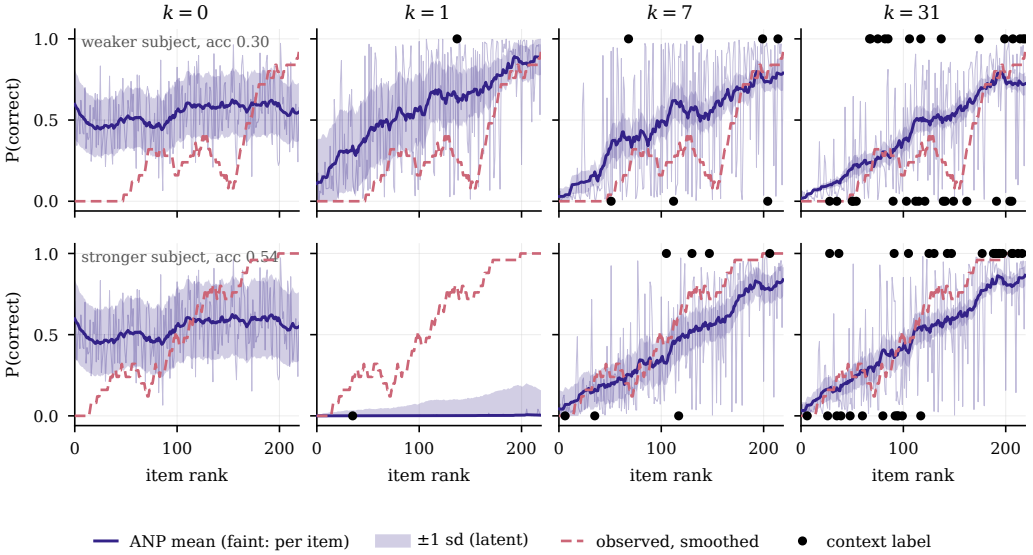
Generalisation: same-subject response-holdout AUC

x: optimiser step (log scale). Values as logged.



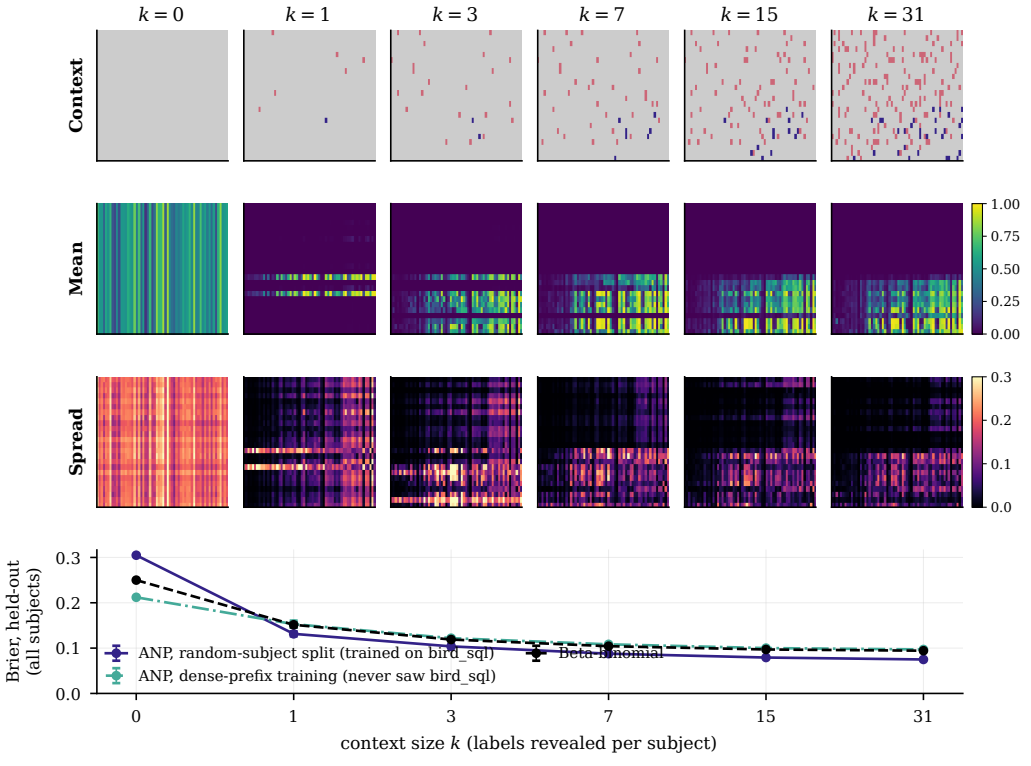
ANP learning one subject from context (in distribution)

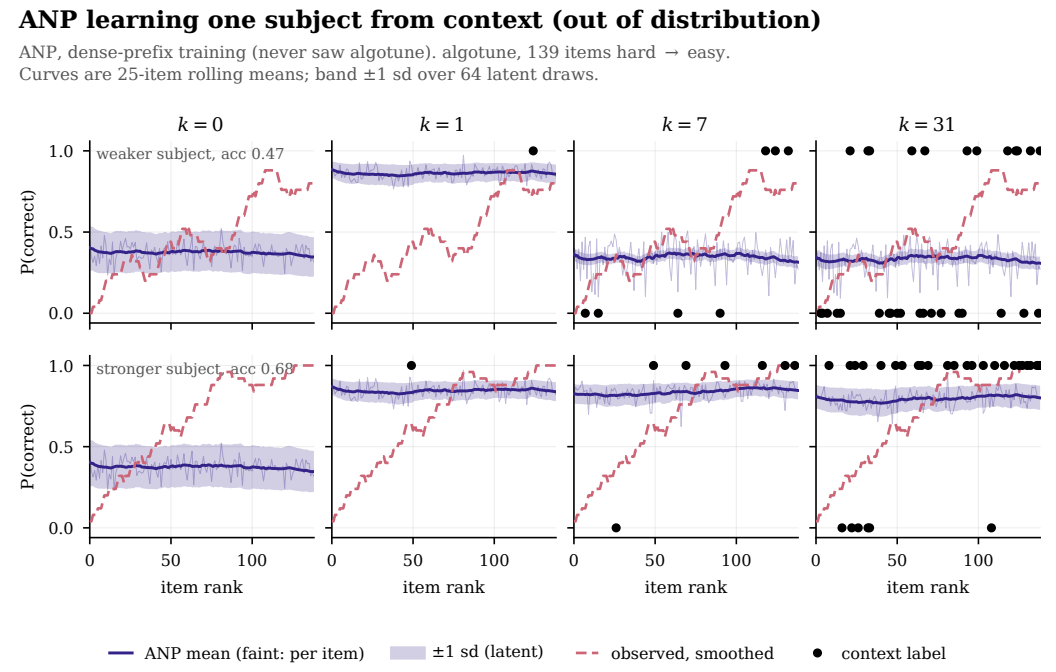
ANP, random-subject split (trained on bird_sql). bird_sql, 220 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.



Response-matrix completion by the ANP (in distribution)

bird_sql: 24 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.




ANP learning one subject from context (out of distribution)
ANP, dense-prefix training (never saw algotune). algotune, 139 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.
k = 0
k = 1
k = 7
k = 31
weaker subject, acc 0.47
stronger subject, acc 0.68
P(correct)
1.0
0.5
0.0
0
50
100
item rank
ANP mean (faint: per item)
±1 sd (latent)
observed, smoothed
context label

Response-matrix completion by the ANP (out of distribution)

algotune: 18 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.

